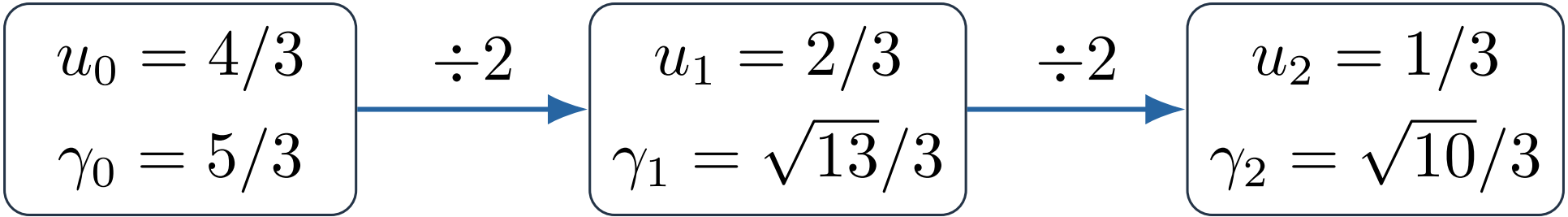
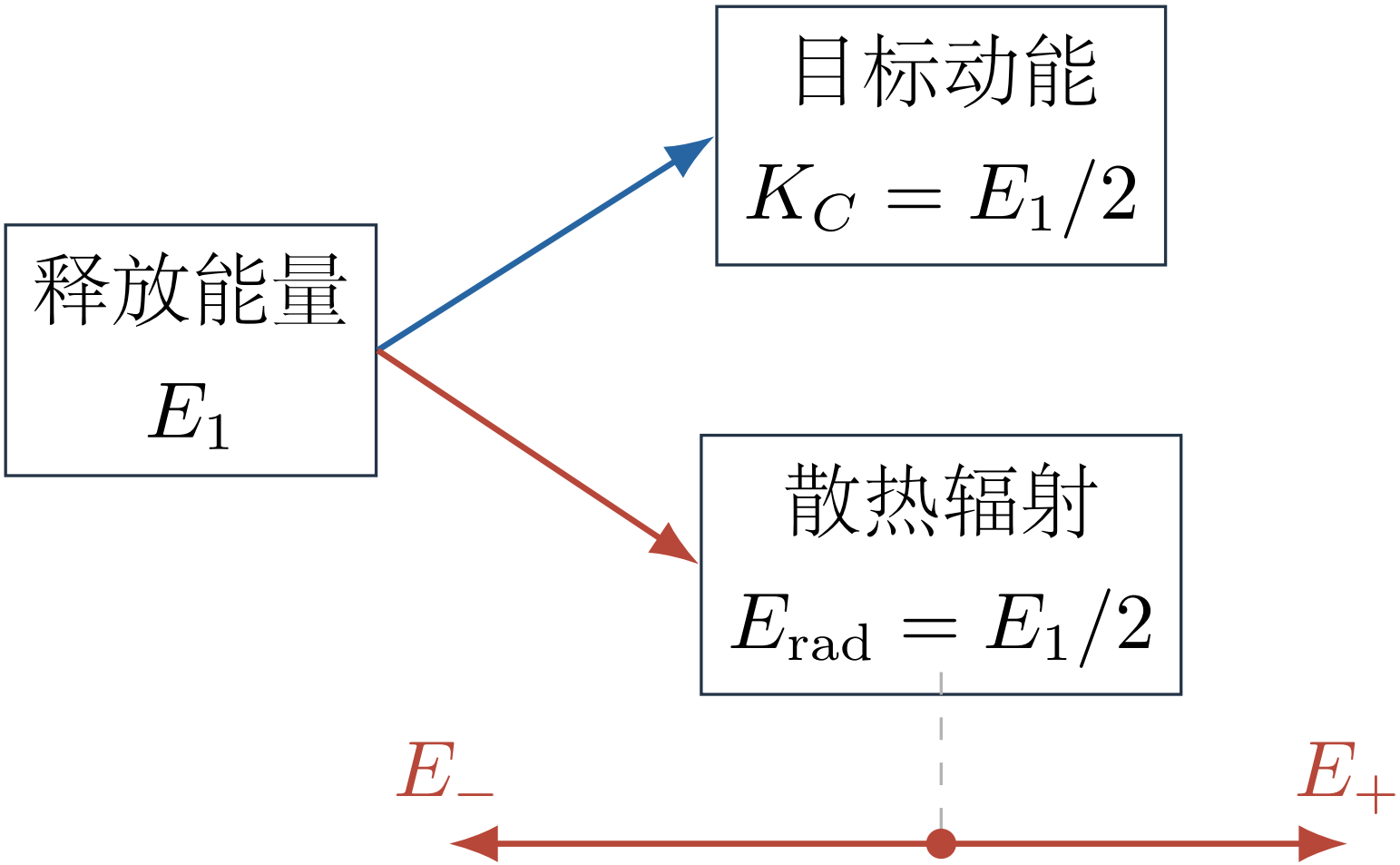
静质量 m_0　　初始静止，静质量 m_0

同一地面参考系；攻击不改变运动者静质量



$$E_j = (\gamma_{j-1} - \gamma_j) m_0 c^2$$



$$E_+ + E_- = E_{\mathrm{rad}}, \quad p_{\mathrm{rad}} = (E_+ - E_-)/c$$

本题 $E_- > E_+$，净辐射动量向负方向